ASSIGNMENT : 01

① Explain the Evolution of Big data from traditional Systems.

Ans: *) DATA GENERATION AND STORAGE :

=> 1970s and Before
=> Mainframes : Basic data storage.
=> primitive and structured data.

*) DATA UTILIZATION :

=> 1980s and 1990s.
=> Relational Databases.
=> Data - Intensive Applications.
=> complex and Relational Data.

*) DATA DRIVEN :

=> 2000s and Beyond.
=> Structured, Unstructured, Multimedia, complex Data.

(2) Compare Structured, Semi-Structured and Unstructured data with Examples.

Ans:- STRUCTURED DATA:

=> Data which is an Organized form.
ie. rows and columns.

=> It can be easily used by a computer program.

=> Relationship exist between entities of data, such as classes and their objects.

=> Data stored in databases is an example for structured data.

Ex:

id	name	age
1	abc	20
2	xyz	25

UNSTRUCTURED DATA:

=> Data which does not conform to data model or it is not in a form which can be used easily by a computer program.

=> About 80% - 90% of an Organization data is in this formal.

Ex: Images. videos. etc.

SEMI-STRUCTURED DATA:

=> Data which does not conform to a data model but has some structure.

=> It is not in a form which can be used easily by a computer program.

=> Example: emails, XML etc.

Ex program: ## Document 1 ##

```
{
  "customes ID" : "1023",
  "name" :
    {
      "first" : "AAA",
      "last" : "BBB",
    },
}
```

③ Define Big Data Analytics, Explain classification of analytics.

Ans: BDA is the process of examining and analyzing large volumes of data with high variety and complexity to discover useful patterns, meaningful information and insights that help in better decision-making.

=> It works with datasets whose volume and variety exceed the storage and Processing capabilities of traditional systems.

CLASSIFICATION:

There are two Schools of thought for classifying analytics.

i) First School of Thought.

ii) Second School of thought.

i) First School of Thought:

According to the first School of thought, analytics is classified into four types:

a) BASIC ANALYTICS:

=> Mainly involves slicing and dicing data to obtain basic business insights.

=> Includes: Historical reporting, Basic visualization.

Ex: preparing a report showing last year's sale.

b) OPERATIONALIZED ANALYTICS:

=> Analytics is integrated into enterprise business process

=> It helps Organizations uses analytical results directly during their regular Operations.

Ex: A System automatically identifying customers who are likely to purchase a product.

c) ADVANCED ANALYTICS:

=> focuses on future Engineering.

=> uses: predictive & prescriptive model.

Ex: predicting future Sales & Suggesting what actions should be taken

d) MONETIZED ANALYTICS :

=> Analytics is used directly to generate business revenue.

=> The analytical results themselves become a source of business value or income.

9i) Second School of Thought :

According to the Second School of thought, analytics is classified into :

a) Analytics 1.0 :

=> Descriptive Analytics.

=> Mainly deals with what happened in the past.

=> Uses traditional data from legacy Systems, ERP, CRM etc.

=> Data is generally small & Structured.

=> Uses relational databases.

=> Key question asked : "what happened?".

b) Analytics 2.0 :

=> predictive analytics.

=> Uses big data and predictive statistics to Understand what is likely to happen in future.

=> Handles large-scale data using technologies Such as Hadoop.

=> Key questions asked : what will happen?, why will it happen?.

c) Analytics 3.0 :

=> prescriptive Analytics.

=> combines descriptive & predictive analytics.

=> Uses both big data & traditional data.

=> Uses machine learning, agile methods and advanced database processing.

=> Key questions asked : Why will it happen?
what should we do?
How can we make it happen?

(4) What is NoSQL ? Explain the different types of NoSQL databases with an example.

Ans: NoSQL Stands for Not Only SQL.

These are :

=> Non-relational database.

=> Open-Source database.

=> Distributed database.

=> They are popular because they Scale horizontally, Handle Structured, Semi-Structured, & Unstructured data.

** Types of NoSQL Databases :

i) Key-value DB : Maintains a large hash table of keys & values.

Ex : Dynamo, Riak.

Key	Value
First Name	Abc
Last Name	XYZ

ii) Document DB : Stores data in collections of documents.

Ex : MongoDB, Couchbase.

```
{
  "Book Name" : "Introduction to AI",
  "publisher" : "abc"
}
```

iii) Column-Oriented DB : Each Storage Block contains data from only one column.

Ex : Cassandra, HBase.

iv) Graph DB : Also called a network Database.

=> A graph stores data in nodes, Edges.

Ex : Neo4j.

(5) Social Media platforms generate massive flows of post, videos, emojis & Hashtags every second, creating enormous volumes of complex information. why is this data Unstructured and what challenges information do traditional methods face in processing it?

Ans: Social Media data is Unstructured because it is generated in different formats such as text, images, videos & audio etc.

a) and it does not follow a fixed table or predefined Schema.

CHALLENGES:

1) Difficult to Store & Organize Such varied data.
2) Traditional DB mainly handle Structured data.
3) Processing huge volume of data is slow.
4) Difficult to Extract meaningful information.
5) Requires large storage.

(6) Give the HDFS commands to perform the following Operations.

i) List directories and file at root.

Command: hdfs dfs -ls /

ii) Create a directory.

Command: hdfs dfs -mkdir / directory-name.

iii) Copy local file to HDFS.

Command: hdfs dfs -put localfile / directory-name.

iv) Display content of HDFS file.

Command: hdfs dfs -cat / directory-name / filename.

v) Remove a directory.

Command: hdfs dfs -rm -r / directory-name.

(7) With a neat diagram, Explain HDFS architecture.

Ans :-

*) Key Components :

1) client Application :

=) The client interacts with HDFS through the Hadoop file System client to read / write data.

2) Name Node :

=) Stores metadata about files.

=) Manages file-to-block mapping.

=) Assigns block to different Datanodes for storage.

3) Data Nodes :

=) Actual storage locations for blocks.

=) Each block is replicated across multiple Datanodes.

⑧ List and Explain the Components of YARN in Hadoop.

Ans:

i) Resource Manager:

=> It is a global resource manager of YARN.

=> It distributes resources among different application.

=> It has two components:

a) Scheduler: Allocates resources to applications.

b) Application Manager: Accepts job Submissions and Manages Application Master.

ii) Node Manager:

=> It is a per-machine daemon.

=> It launches and manages application containers.

=> It monitors such as CPU, memory, disk & network.

iii) Application Master:

=> It is application specific.

=> It works with the NodeManager to Execute and monitor application tasks.

④ Compare Mapper & Reducer functions with Example.

Ans: MAPPER:

=> Converts Input key-value pairs into Intermediate key-value pairs.
=> performs filtering and transformation of Input data.
=> Works on the Input data Split.
=> Output is Intermediate key-value pairs

Ex: (Hello, 1) → (Hello, 1)

REDUCER:

=> Reduces Intermediate values having the same key into a same key into a Smaller set of values.
=> Performs aggregation and processing of grouped data.
=> Works on the Output produced by the mapper after Shuffle & Sort.
=> Output is the final key-value pairs.

Ex: (Hello, [1, 1]) → (Hello, 3)

Example - Word count.

Input: Hello World World

Mapper Output: (Hello, 1)
(World, 1)
(World, 1)

Reducer Output: (Hello, 1)
(World, 2)

(16) What is MongoDB? Explain Creation of database, dropping of database with Example.

Ans:- MongoDB:

MongoDB is a NOSQL, document Orented DB that Stores data in JSON like documents (BSON).

* Creating Database:

Syntax: use database_name

Example: To creat a database named myDB, use:

use myDB

o/p: Switched to db myDB

* Verifying to current DB:

db

o/p: myDB

* Listing All DB's:

Show dbs.

o/p: local 0.078 GB
test 0.078 GB

* Drop Database:

Syntax: db.dropDatabase();

* Steps to Drop DB:

=> first Switch to DB you want to drop.
=> use myDB;
=> Then Execute Drop Command.

(11) List and Explain the commonly used terms in RDBMS and mongoDB, compare their corresponding terms with suitable Example.

Ans:- STRUCTURAL DIFFERENCE:

RDBMS	MongoDB	Example
Database	Database	Ex: StudentDB can contain Student Information.
Table	Collection	RDBMS → Tables. MongoDB → collections contains Documents
Row	Document	RDBMS → Stores one Students as row MongoDB → { name : "A", age : 20 },
Column	Field	MongoDB has a name field inside a document.
Primary Key	-id field	RDBMS → Student_ID as primary key MongoDB automatically provides unique ID.

RELATIONSHIP & JOINS:

Foreign key	Reference	Student document can store the reference to Dept document
JOIN Operation	Embeddings	Related data can be stored together using Embeddings.

INDEXING & QUERYING:

Index	Index	used to make data Searching faster
SQL query	FIND query	SQL uses SELECT; MongoDB uses find().
Aggregate functions	Aggregate pipeline	Used for operations such as SUM, AVG etc.

TRANSACTIONS & CONSISTENCY:

ACID Transaction	Multi-document Transaction	Ensure multiple operations are completed reliably
Commit & Rollback	Session-based Transaction	Commit saves changes, Rollback cancels changes

PERFORMACE & SCALING:

Vertical	Horizontal	Increase Server power vs add more Servers
Partitioning	Sharding	Divide data into Smaller parts
Replication	Replica Sets	Keeps copies of data on multiple Servers

BACKUP & RECOVERY :

Backup & Restore	mongodump	used to Save and restore data
Point - In - Time	Oplog Replay	Restore data to a Specific time.

DATA INTEGRITY & CONSTRAINTS :

constraint	Schema Validation	Ensure vaild data is stored.
Triggers	change Streams	Detects changes in Data.

SECURITY & AUTHENTICATION :

uses Roles & permissions	RBAC	controls what users can access
Authentication	SCRAM	Verify user Identity
Encryption	Encryption at REST & In Transit	Protects Data.